

ANKARA ESENBGA, Turkey

WMO: 171280

Lat: 40.128N

Lon: 32.995E

Elev: 3125

StdP: 13.11

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.3	13.9	3.2	7.3	10.7	7.9	9.3	17.7	22.3	37.6	19.2	36.8	2.4	30	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	26.9	93.1	62.6	89.8	62.0	87.2	61.6	66.3	84.4	64.7	82.9	63.3	81.4	8.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.6	85.6	73.2	57.7	80.0	71.9	56.5	76.4	70.8	32.8	83.9	31.4	82.8	30.3	81.4	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	16.8	14.8	DB	1.7	97.8	7.1	2.8	-3.5	99.8	-7.7	101.5	-11.7	103.1	-16.9	105.1	(4)
(5)			WB	1.0	68.2	7.0	2.8	-4.1	70.2	-8.2	71.8	-12.1	73.4	-17.2	75.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	31.1	34.5	41.1	49.7	58.2	65.4	71.9	72.4	63.7	53.7	42.4	34.3		
	DBStd	15.74	8.63	8.40	7.24	6.77	5.70	5.09	4.83	4.85	6.17	6.57	6.96	7.37		
	HDD50	2160	587	436	284	83	6	0	0	0	0	38	240	486		
	HDD65	5466	1052	855	740	459	219	55	5	5	94	353	678	951		
	CDD50	2756	0	1	9	75	261	461	678	694	413	152	11	1		
	CDD65	586	0	0	0	1	8	66	217	235	57	2	0	0		
	CDH74	8111	0	0	1	37	272	1059	2824	2838	970	110	0	0		
	CDH80	3121	0	0	0	2	42	309	1219	1219	312	18	0	0		
(14) Wind	WSAvg	5.8	4.7	5.4	6.4	6.1	5.8	6.4	7.5	7.4	5.9	5.0	4.2	4.4		
(15) Precipitation	PrecAvg	16.6	2.0	1.5	1.5	1.8	2.1	1.4	0.6	0.4	0.7	1.1	1.3	2.2		
	PrecMax	22.2	4.1	3.5	3.7	3.9	4.4	3.7	2.5	1.2	2.2	3.1	4.4	5.2		
	PrecMin	10.5	0.2	0.3	0.4	0.2	0.7	0.4	0.0	0.0	0.0	0.0	0.1	0.2		
	PrecStd	2.5	1.2	0.9	0.8	1.0	0.9	0.8	0.6	0.4	0.7	0.8	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	60.7	69.5	78.7	84.3	90.2	97.2	96.5	91.5	82.5	68.2	55.6		
		MCWB	44.4	46.5	49.5	54.8	59.2	62.1	63.5	62.6	59.7	58.0	51.5	47.0		
	2%	DB	48.6	55.2	63.6	73.4	80.5	86.4	93.5	93.0	87.4	76.9	64.2	51.4		
		MCWB	42.1	44.3	47.6	53.1	58.5	61.4	63.2	62.6	59.4	55.8	50.6	44.4		
	5%	DB	45.6	51.5	59.3	69.4	77.0	83.8	90.0	89.9	83.8	72.9	59.7	48.2		
		MCWB	40.6	42.8	45.8	52.0	57.2	60.8	62.8	61.9	58.6	54.6	48.2	43.0		
	10%	DB	42.5	47.8	55.4	64.7	73.4	80.5	86.7	87.2	79.9	68.3	55.6	45.0		
		MCWB	39.0	41.3	44.2	49.9	56.3	59.6	62.1	61.8	57.9	53.2	46.4	41.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	48.4	51.3	57.6	62.6	66.5	69.0	68.8	64.1	60.0	54.1	47.9		
		MCDB	52.4	56.7	65.3	72.8	77.0	81.8	88.3	88.3	82.2	75.7	64.4	53.8		
	2%	WB	43.0	45.5	48.7	54.6	60.3	63.8	66.6	66.0	62.2	57.4	51.4	45.6		
		MCDB	47.8	53.2	60.7	69.8	75.6	80.5	85.7	83.8	79.5	70.7	61.4	50.1		
	5%	WB	40.8	43.4	46.8	52.7	58.6	62.1	65.0	64.5	60.5	55.8	49.1	43.6		
		MCDB	44.4	50.1	57.5	66.4	73.6	78.6	83.8	82.7	77.9	68.6	57.8	47.5		
	10%	WB	38.9	41.3	45.0	50.9	57.0	60.9	63.4	63.2	59.0	54.5	47.3	41.6		
		MCDB	42.0	46.9	54.3	62.9	70.8	76.9	82.2	81.5	76.8	66.3	54.6	44.7		
(35) Mean Daily Temperature Range	MDBR		14.3	18.1	21.3	23.5	24.8	26.2	27.7	26.9	27.5	24.7	22.8	15.0		
	5% DB	MCDBR	19.3	24.9	29.6	31.4	31.0	31.5	32.0	30.8	33.4	32.5	30.2	19.8		
		MCWBR	13.5	15.6	16.3	15.1	13.7	12.1	11.4	10.6	14.0	16.3	18.2	13.6		
	5% WB	MCDBR	17.2	22.6	26.0	27.4	27.6	28.1	29.1	27.3	28.3	26.6	26.1	16.8		
		MCWBR	12.4	14.5	14.9	14.3	13.4	12.0	11.7	10.7	12.9	14.0	16.2	11.9		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.328	0.372	0.409	0.387	0.368	0.365	0.364	0.344	0.344	0.313	0.303		
	taud		2.373	2.323	2.236	2.163	2.292	2.369	2.376	2.360	2.414	2.406	2.460	2.436		
	Ebn at Noon		272	280	278	274	283	287	287	285	271	267	263	263		
	Edh at Noon		27	33	40	46	41	38	38	37	33	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		591	908	1269	1625	1931	2202	2346	2090	1640	1118	771	521		
	RadStd		75	138	116	152	148	189	102	94	116	83	76	69		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page